

TOPIC 11. COMPOSITE MATERIALS (Session I)

Concept of composite materials

Constituents of composite materials

Classification of composite materials

Particle-reinforced composites

- Large particles
- Fine Particles

Fiber-reinforced composites

- Glass fibers
- Carbon fibers
- Aramid Fibres
- Woven fabrics
- Other fibers

Structural composites

Mechanical properties of different reinforcement materials



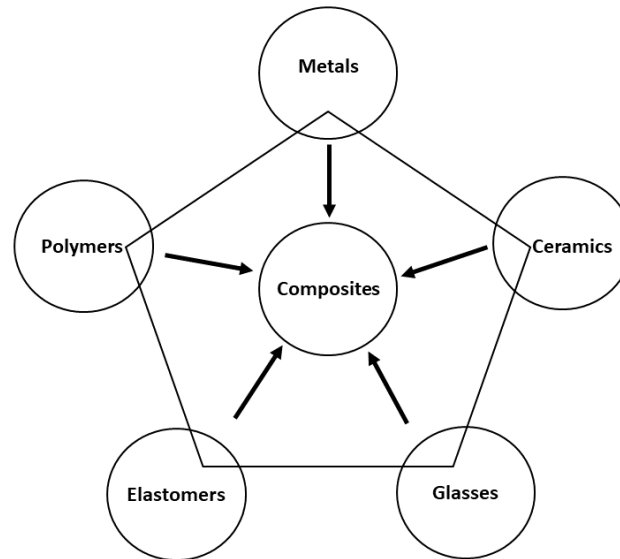
Concept of composite materials

What is a composite material?

“**Mixture** of two or more **distinct materials** with significantly different physical or chemical properties which remain separate with one **clear interface**, and whose constituents retain their chemical identity”

Usually a ***synergistic effect*** in one or more properties is observed.

- **SYNERGISM**: The overall performance of the material should be better than the separate components for a particular applications. Properties *vary continuously* over a range.



Concept of composite materials

COMPOSITES IN NATURE



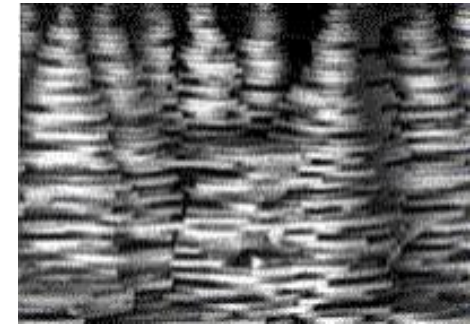
Bettye L. Smith (UCSB)

Abalone shell:

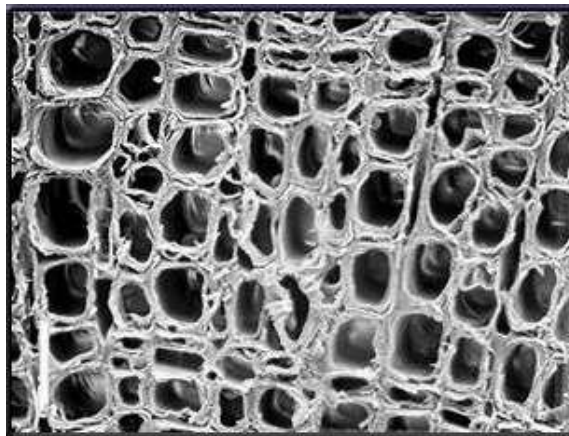
CaCO_3

+ 3% organic material

>3000* stronger than calcite



SEM



Juniperrus ashei wood

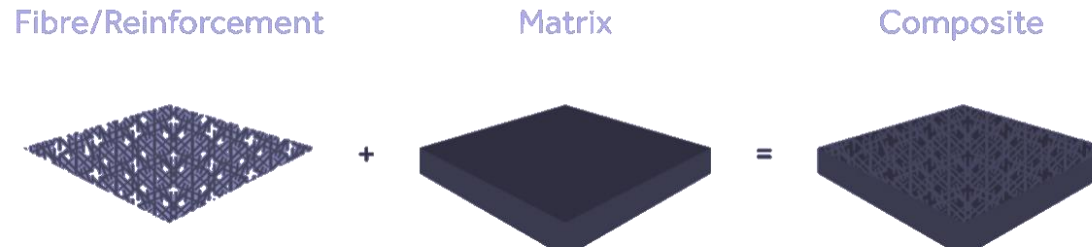
Wood

cellulose-filaments in a matrix of lignin and hemicellulose

growth rings form a layered composite

perpendicular to the growth rings are radially oriented ribbon-like structures : rays which provide a radial stiffening and reinforcement

Constituents of composite materials



Composite: A multi-phase material that exhibits a significant proportion of the properties of the constituent phases such that a better combination of desirable properties is realized.

Matrix: The **continuous phase** that surrounds the other constituents and binds them into a composite material.

Dispersed Phase/ Reinforcement: The **discontinuous**, or **discrete phase** that adds reinforcement to the composite.

- **Particle:** A relatively short globular or spherical dispersed phase.
- **Fiber:** A reinforcement, typically cylindrical, with a diameter smaller than its length

Interface: The planar region that separates two components in any material.

- ⇒ Both the fiber and matrix can be a **polymer**, a **metal** or a **ceramic**.
- ⇒ The properties of composites are a function of the properties of the **constituent phases**, their relative **amounts**, and the **geometry** of the dispersed phase.

Constituents of composite materials

Properties of composite materials for different matrices and the reinforcements

Matrix	Reinforcement material	Properties
Metal	Metal fibers, ceramic, carbon, glass	Mechanical strength to high temp Electric conduction. ↑ thermal stability
Ceramic	Particles and metallic fibers and ceramics	Chemical and thermal resistance to temp. ↑
Glass	Glass and ceramic particles	Mechanical strength and chemical resistance to temp. ↑ thermal stability
Organic	Carbon, glass and organic fibers	Chemical and electrical, and erosion resistance, flexibility

In order to design a composite material, the properties to consider are:

- **For metallic components (creep) and ceramics (fracture toughness) :** physical properties (thermal, electrical, optical....) and mechanical properties (E, Rm, fracture toughness.....)
- **For polymer components:** Physical and mechanical properties and consideration of the absorption and transmission of water.

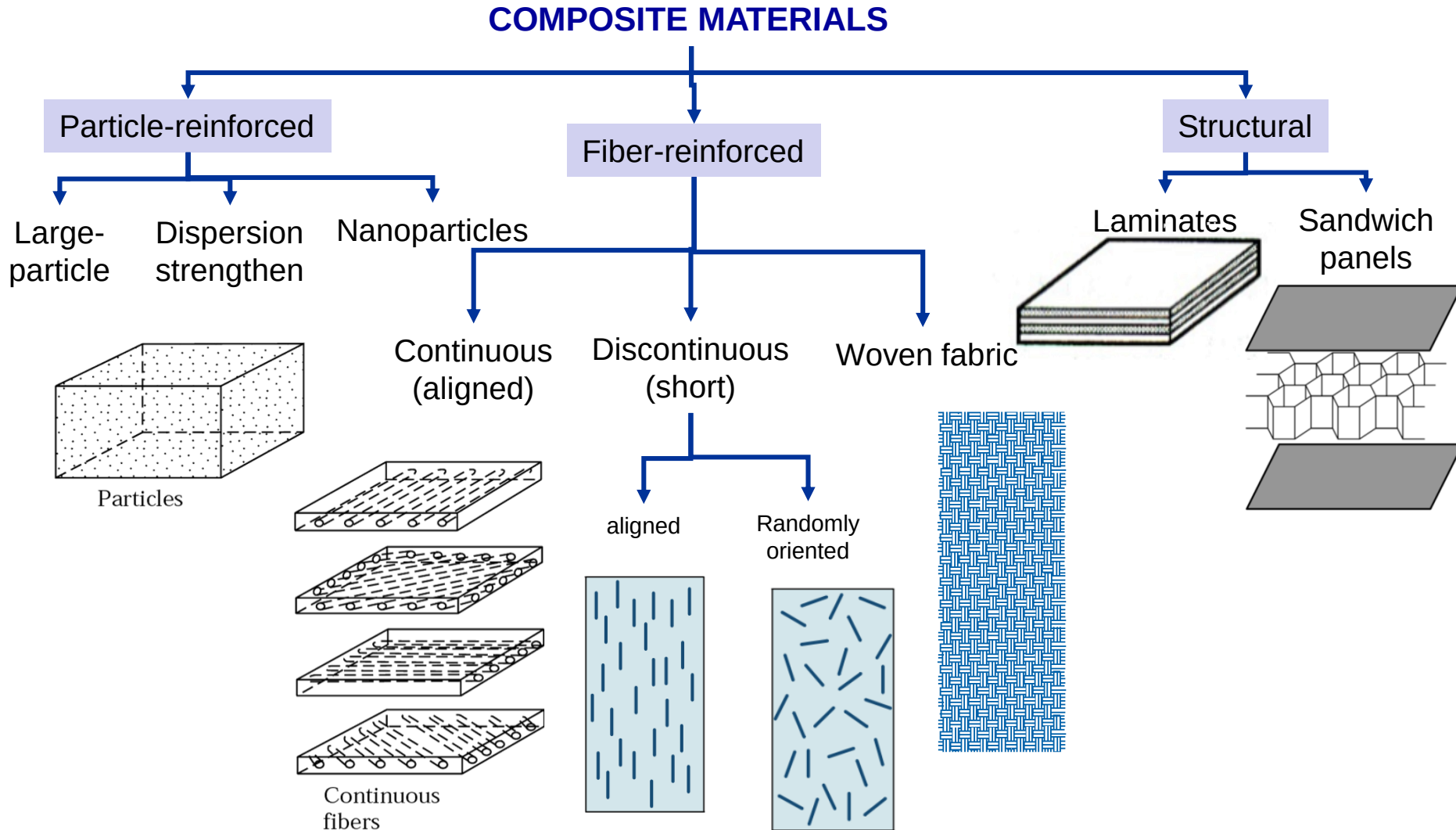
Classification of composite materials

According to the **matrix**:



Classification of composite materials

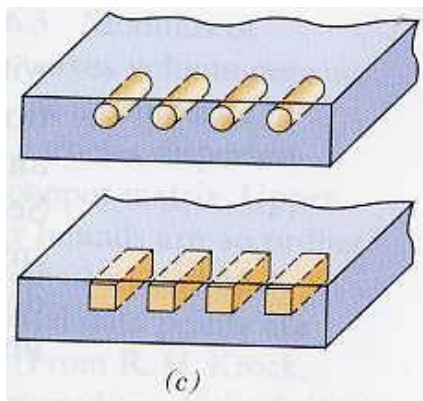
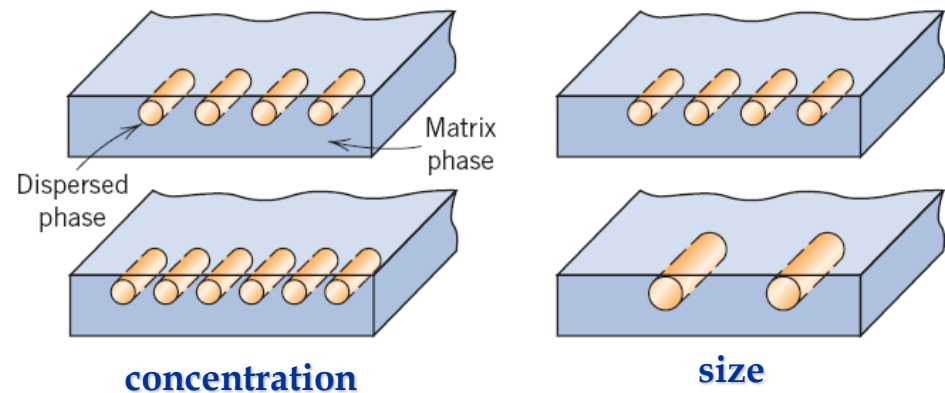
According to the **reinforcement**:



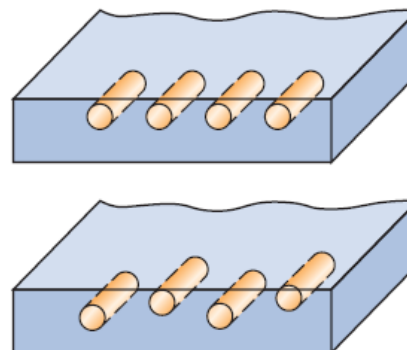
Classification of composite materials: Reinforcements

REINFORCEMENTS

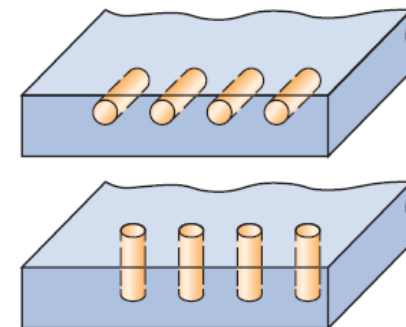
- Particles (*dispersion strengthened - large particles - nanoparticles*)
- Fibers (*discontinuous - short or continuous – aligned- woven*)
- Structural (*laminates and sandwich structures*)
- Factors that influence physical and mechanical properties of composites:
size, distribution and particle content



shape



distribution

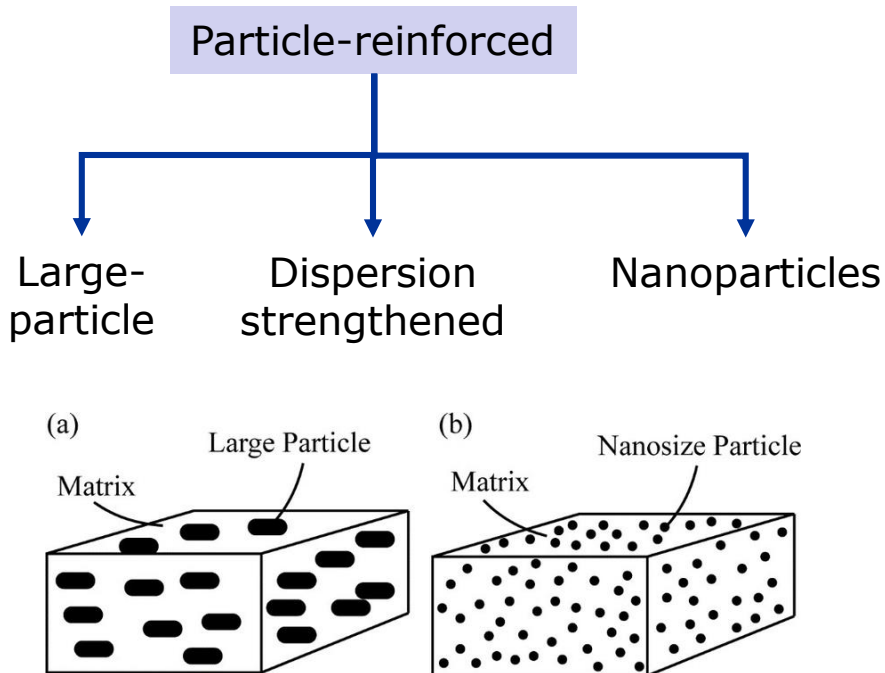


orientation

Particle-Reinforced Composite Materials

PARTICLE-REINFORCED COMPOSITE MATERIALS

- Mechanical properties depend on the reinforcement, manufacturing and subsequent treatments
- Most used metallic matrixes are Al, Mg, Ti y Ni
- Polymeric matrixes are reinforced to improve their mechanical strength and abrasion resistance and electrical conductivity (nanoparticles)



Advantages of Particle Reinforcement:

- ✓ Low cost
- ✓ High stiffness and strength (inorganic particles)
- ✓ Wear resistance
- ✓ Ease of processing

Particle-Reinforced Composite Materials

REINFORCEMENT TYPE: PARTICLES

Large Particles

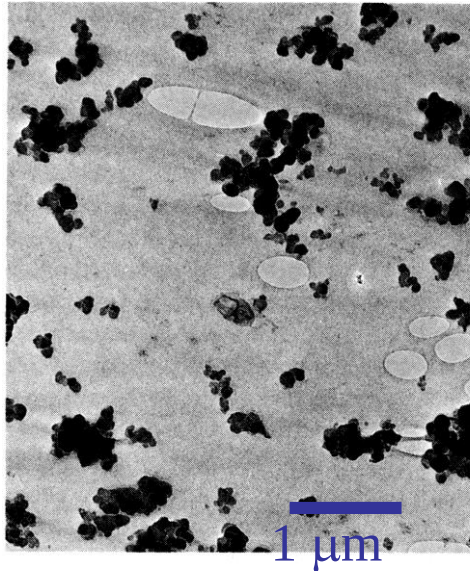
- Particle size= 1-50 μm
 - Concentration= 15-40% by volume
 - The reinforcement-matrix interaction is treated at the **macro-meso** scale
 - Reinforcement is usually harder and stiffer than the matrix
 - Matrix phase transfers applied stress to particle. The reinforcement bears or helps to bear the load.
 - The degree of the mechanical strengthening depends on the bond strength of the matrix-particle interface
-
- **Metals and ceramics** $\Rightarrow$ particles are added to improve toughness and mechanical resistance
 - **Plastics** $\Rightarrow$ particles are added as a filling to improve properties (carbon black - soot, elastomers) or reduce cost (CaCO_3 , clays, hollow glass spheres...)

Particle-Reinforced Composite Materials

REINFORCEMENT TYPE: PARTICLES

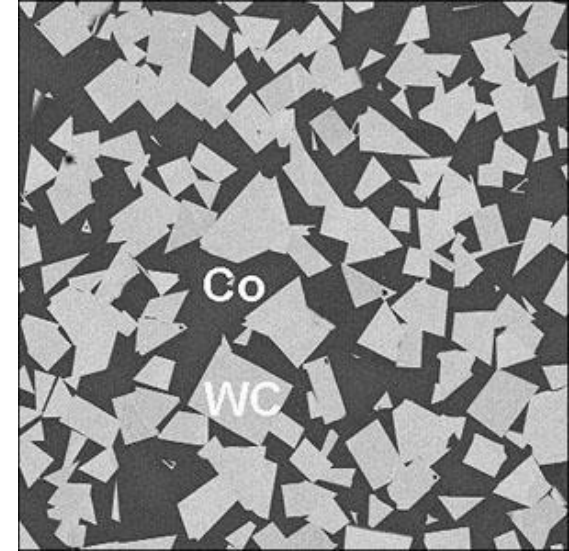
Large Particles

EXAMPLES



Carbon Black particle reinforcement in Styrene-Butadiene synthetic rubber for car tire application.

Vulcanized rubber is a material that undergoes a chemical process known as vulcanization. This process involves mixing natural rubber with additives such as sulfur and other curatives. Also called reinforced rubber. Obtained by strengthening with carbon-black particles. **Used in auto tires**



WC–Co cemented carbide.

Cemented carbides or cermet. Are composites of ceramic particles (strong, brittle) in a metal matrix (soft, ductile) that enhances toughness. For instance, tungsten carbide or titanium carbide ceramics in Co or Ni. They are used for cutting tools for hardened steels.

Particle-Reinforced Composite Materials

REINFORCEMENT TYPE: PARTICLES

Fine Particles

- **Dispersed phase:** Uniformly dispersed fine, hard and inert particle is used for reinforcement
- Particle size $< 0.1 \mu\text{m}$
 - ⇒ *Impede dislocation movement thereby promoting hardening ($\uparrow$ hardness, $\uparrow E$, $\uparrow R$)*
- Volume fraction is between 15-40%.
- Stronger than pure metal.
- Can be metallic, nonmetallic, intermetallic.
- Dispersoids are usually carbides, oxides, borides.
- Shape can be round, disc, needle(max strength).
- **Continuous phase:** matrix continues to carry the load

Particle-Reinforced Composite Materials

REINFORCEMENT TYPE: PARTICLES

Fine Particles

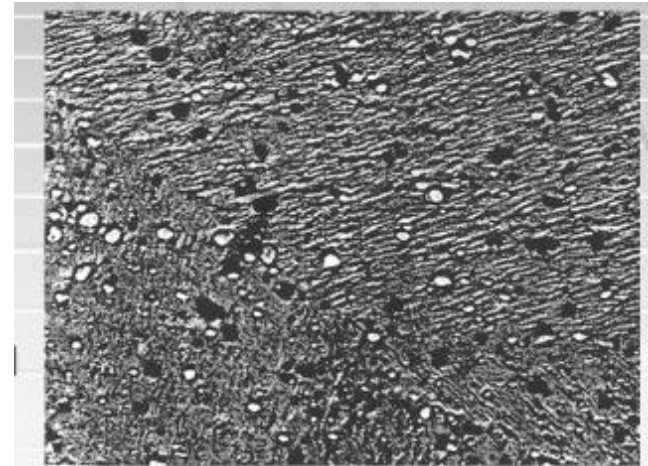
EXAMPLES

➤ **SAP (Sintered Aluminum powder/product)**

Fine Al_2O_3 particles are dispersed in pure aluminum matrix

➤ **TD-Nickel (Thoria Dispersed Nickel)**

Powders of metallic Th and Ni are ball milled, compacted and sintered in air –Th reacts to form ThO_2



Electron micrograph of TD-Ni with 300 nm diameter ThO_2 particles (X2000)

Fiber-Reinforced Composite Materials

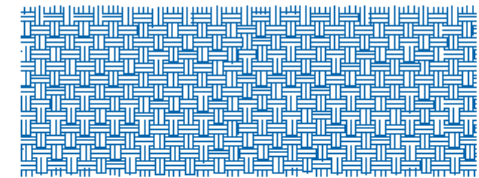
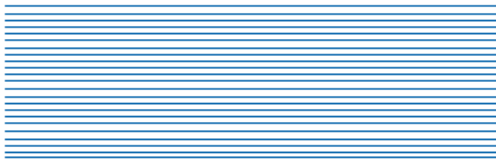
Reinforcement Type: Fibers

CLASSIFIED ACCORDING TO SHAPE

Continuous fibers
(aligned)

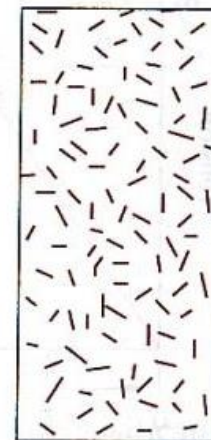
discontinuous or short fibers (aligned and
randomly oriented)

Woven fibers



Longitudinal direction

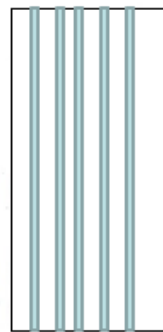
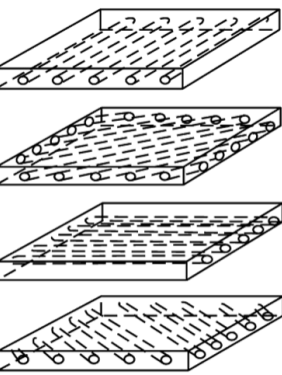
Transversal
direction



(b)

(c)

- a) continuous and aligned fibers
- b) discontinuous and aligned fibers
- c) discontinuous and randomly oriented fibers



Continuous
fibers

Fiber-Reinforced Composite Materials

FIBER-REINFORCED COMPOSITE MATERIALS

Fiber types:

- * **Glass fibres (GFRP)**
 - * **Carbon fibres (CFRP)**
 - * **Aramide (AFRP)**
- } for reinforcing polymers
- * Other fibers: boron, SiC, Al₂O₃

Matrix types:

- * Polymeric:
 - Thermosetting (usually epoxy - for continuous fibers)
 - Termoplastic
- * Metal and ceramic (less used)

Two primary factors limit their applications:

- ⇒ **Absence of plasticity** (critical in mechanical joints)
- ⇒ **Low damage tolerance** ⇒ small accidental impacts can produce delaminations that instigate a loss of $R_{\text{mechanical}}$

Reinforcement: Fibers

FIBER-REINFORCED COMPOSITE MATERIALS

FIBER FUNCTION

- Carry most of the load applied to the composite

Need
GOOD INTERFACE BOND
FIBER $\Leftrightarrow$ MATRIX

MATRIX FUNCTIONS

- Transfer load from into and out of the fibers
- Protects the fiber from the environment (chemical degradation)
- Separates the fibres to prevent failure of adjacent fibers if one fails (Avoid crack propagation)
- Maintain the orientation of the fibers
- Form the shape of the component

High Performance Composite Materials

Strong, stiff, and brittle
fibers



Soft and ductile matrix



*High fatigue strength,
stiffness in relation to its
weight*

ULTIMATELY: achieve high strength and stiffness along with low density (High Specific Strength and High Specific Modulus)

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

GLASS FIBERS

- **Composition:** Base of SiO_2 (50-70%) + Oxides Ca, Al, B, Na, Mg and K
- **Properties:** non-combustible; good chemical, biological and thermal resistance ($T_m \uparrow, \alpha \downarrow$); electrically and thermally insulating ($k \downarrow$); low coefficient of thermal expansion ($\alpha \downarrow$); low cost; amorphous and isotropic.

Glass fibers exhibit:

- Higher tensile strength
- Lower modulus (however more elongation)
- Higher density

When compared with **C** and **aramid** fibers

But due to their **low cost** and **versatility, easy processing**, they are used more to reinforce polymers

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

GLASS FIBERS

Fiberglass properties							
Type of glass	ρ_{relative}	σ_{tensile} (MPa)	E (GPa)	$\alpha \times 10^{-6}$ (K)	ϵ (a 20 °C y 1 MHz)	T_m (°C)	For applications that require
E	2,58	3450	72,5	5,0	6,3	1065	Good electrical properties and dimensional stability (circuit boards)
A	2,50	3040	69,0	8,6	6,9	996	Chemical resistance
ECR	2,62	3625	72,5	5,0	6,5	1204	Good electrical properties and chemical resistance
S	2,48	4590	86,0	5,6	5,1	1454	Tensile strength and thermal stability (aerospace and aeronautic industries)

E Glass: Ca Aluminoborosilicate cheapest and highest moisture resistance

A Glass: Rich in alkali

S Glass: Mg Aluminosilicate without B

Common polymeric matrixes:

Thermoplastics: Nylon 66, Polycarbonate and Polystyrene

Thermoplastics: Epoxy, polyesters, phenolic, silicon

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

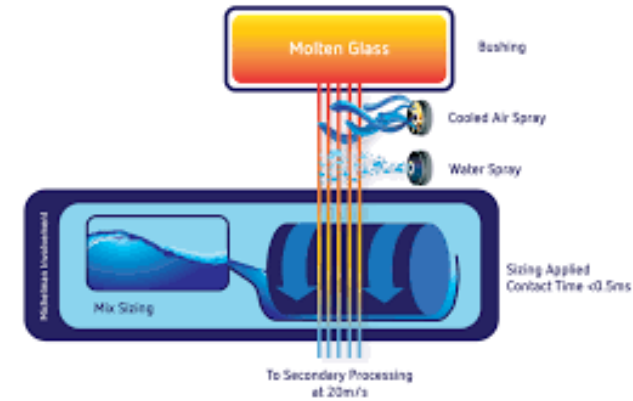
GLASS FIBERS

- Strength of glass fibres is determined by: **damage** during processing, and the **adsorption** of water over their surface

Sizing: Protective coating applied to glass fibers. Sizing normally contains a **film-forming polymer**, such as polyvinyl acetate, and a **coupling agent** (organosilane, titanate or chromate).

Role of sizings

- **binds** the filaments ($d = 10\text{-}20\text{ }\mu\text{m}$) together for ease processing,
- **lubricates** fibres so they can withstand abrasion,
- imparts **anti-static** properties
- provides some **chemical functionalities** to the surface to improve interfacial bond strength.



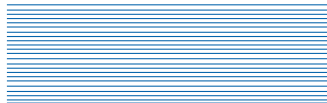
Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

GLASS FIBERS

Typical configurations for fiberglass reinforced composite materials:

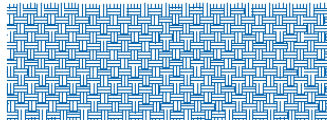
- a) Continuous fibers
- b) Discontinuous
- c) Woven fiber (dry or prepeg)



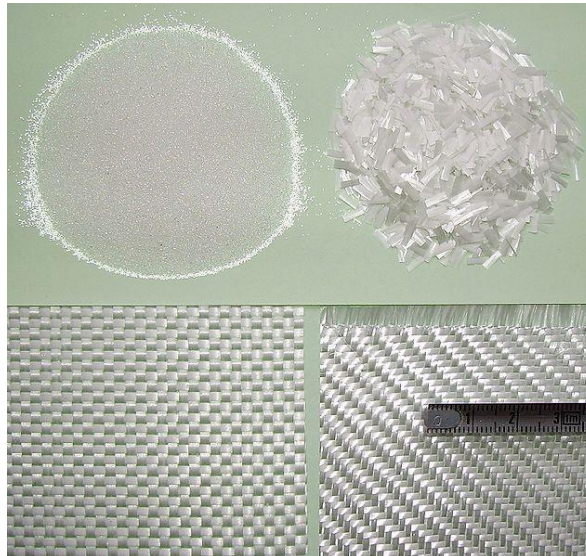
(a)



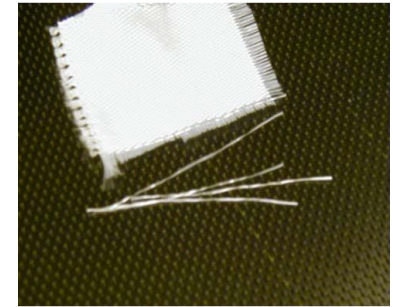
(b)



(c)



Glass fibers



Glass fabric



Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

▪ Advanced composites for aerospace and aeronautic fibers.

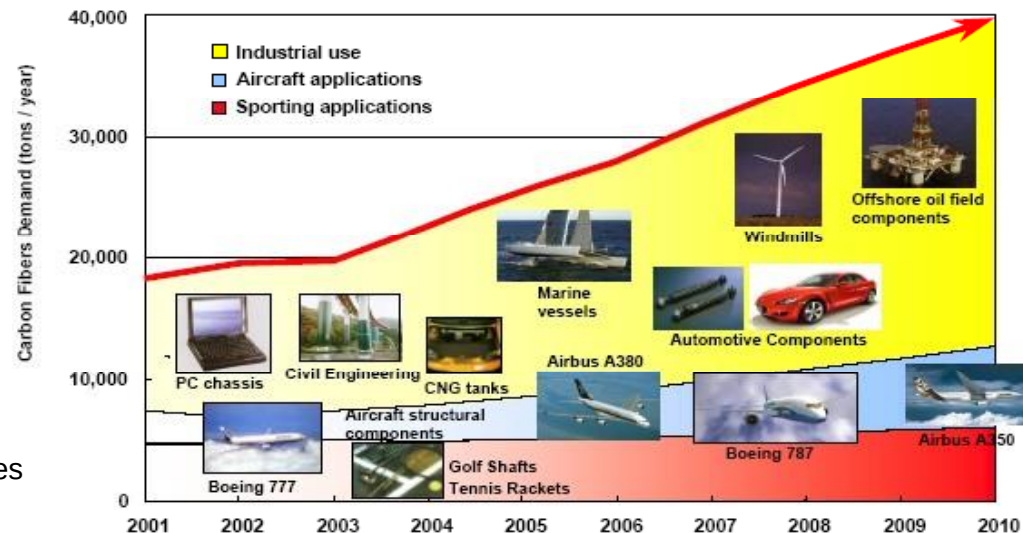
▪ **Very good thermal and physical properties**

- High mechanical strength, stiffness and low weight
- Low CTE and high thermal and electrical conductivity

Combined with the $CTE_{matrix} \Rightarrow$ CM with a $CTE \sim 0$. This minimizes the thermal fatigue strength.

▪ **Carbon fibers** in composites with plastic resins (i.e.: epoxy) good combination of high mechanical strength, stiffness and low weight

▪ Low cost: *sport equipment manufacturing, industrial and commercial products (70's $\approx$ 220 \$/kg and 80's $\approx$ 9\$/kg)*



Evolution of the applications of carbon fiber during the period 2001-2010 source: M. Kamiura. Toray Carbon Fiber Composites Materials Businesses. Toray IR Seminal-No.7. June 6, 2005)

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon fibres: IUPAC classification (according to mechanical properties)

Ultra high elastic modulus type (UHM) :

Tensile elastic modulus: 600 GPa or higher;
Tensile strength: 2500 MPa or higher

High elastic modulus type (HM):

Tensile elastic modulus: 350-600 GPa; Tensile strength: 2500 MPa or higher

Intermediate elastic modulus type (IM):

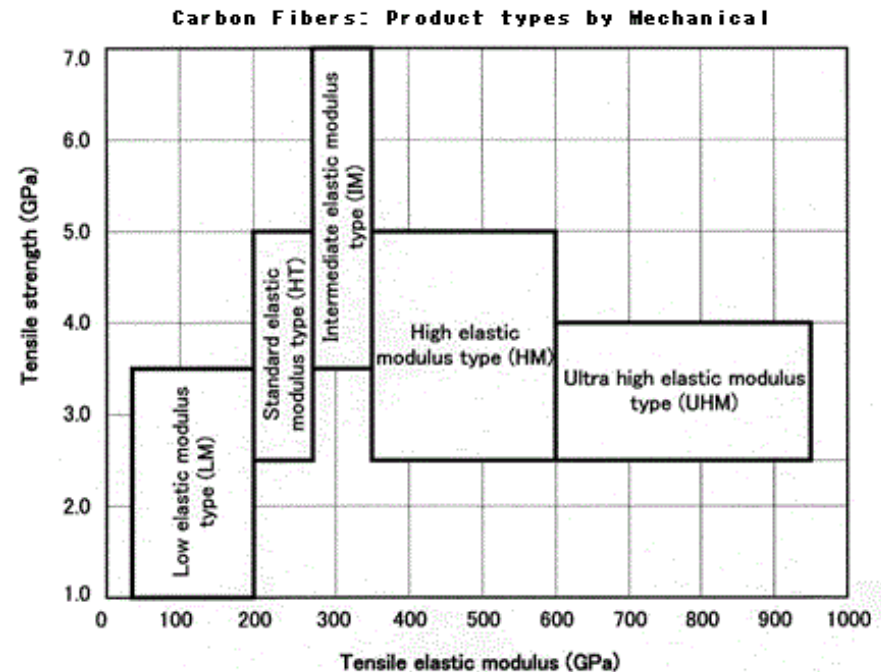
Tensile elastic modulus: 280-350 GPa ; Tensile strength: 3,500 MPa or higher

Standard elastic modulus type (HT):

Tensile elastic modulus: 200-280 GPa ; Tensile strength: approximately 2500 MPa or higher

Low elastic modulus type (LM):

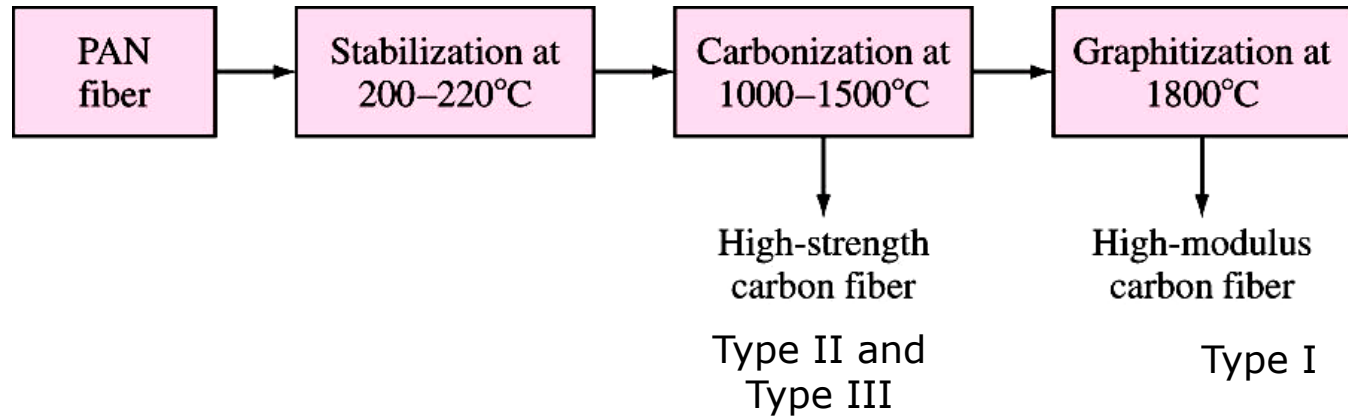
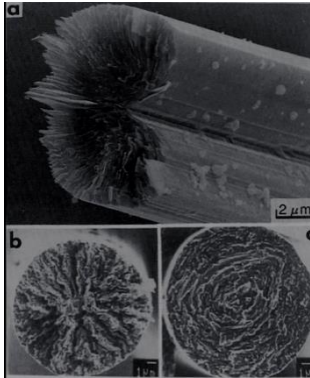
Tensile elastic modulus: 200 GPa or lower ;
Tensile strength: 3500 MPa or lower



Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

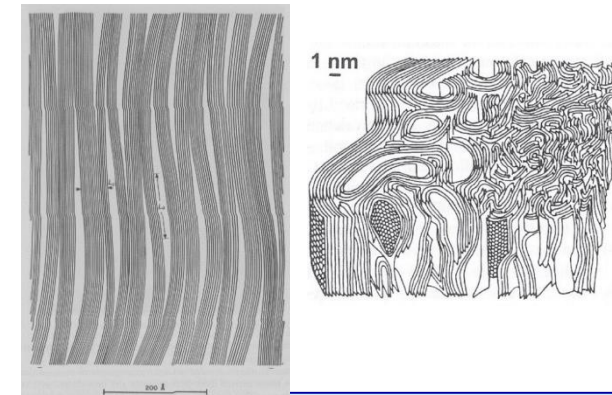


Manufactured from organic precursors:

- **Polyacrylonitrile** (PAN –based fibers)
- **Pitch** (Pitch-based fibers) (oil/coal precursor)

1. PRODUCTION OF ACRYLIC (PAN) OR PITCH FIBER (SPINNING)
2. STABILIZATION OR OXIDATION: Stretching (200-300°C): fibrillar network. Pass the fiber bobbin or thread through an oxidizing furnace in air
3. CARBONIZATION: The fibers are heat treated at 1000-1500°C in inert atmosphere (⇒ Removal O,H,N) and under longitudinal tension
4. GRAPHITIZATION: heat treated at $T > 1800^\circ\text{C}$ in inert atmosphere . Degree of orientation increased: ↑E and strength

C fiber structure with the hexagonal planes oriented in the direction of the fiber orientation



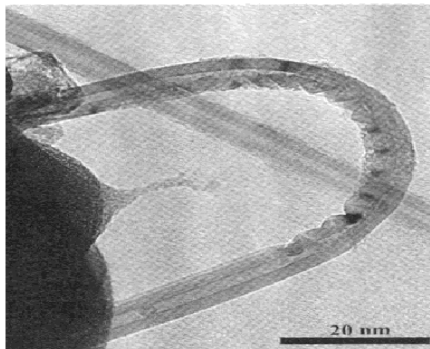
Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon fibres: classification according to heat treatment temperature

- Type I (or HTT): High degree of orientation (high modulus), High temperature heat treatment ($> 2000^{\circ}\text{C}$)
- Type II (or ITT): less degree of orientation but high strength, Intermediate temperature ($\approx 1500^{\circ}\text{C}$)
- Type III (LTT): random orientation (low strength and modulus). Low temperature ($< 1000^{\circ}\text{C}$)



- nanotubes:
promising reinforcements

Fiber-Reinforced Composite Materials

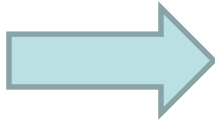
FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon Fiber Products



Individual
filaments
(5-20 μm)

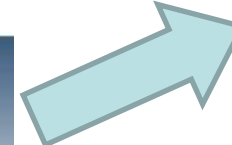


Bundles or tows

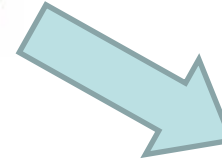
- 1K (1000 filaments)
- 3K (3000 filaments)
- 6K (6000 filaments)
- 12K (12000 filaments)
- 24K (24000 filaments)
- $\uparrow\text{K} \downarrow\text{cost}$



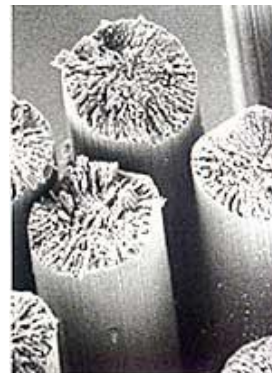
Coil



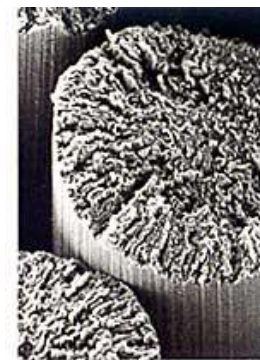
Unidirectional tape



Fabric



X1,500



X3,000



X20,000

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon Fiber Products

Filament



Tow



Staple Yarn



Woven fabric



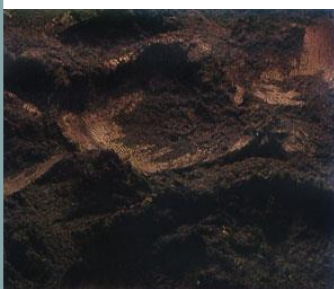
Braid



Chopped fiber



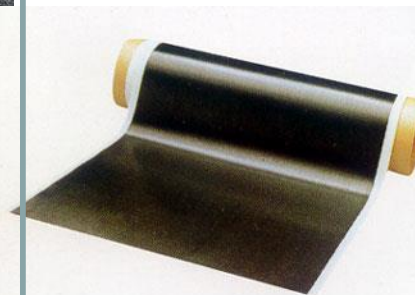
Milled



Felt.Mat



Prepreg



Compound



Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon Fiber Products and Usage

Types	Specifications	Major Usage
Fillament	A yarn constituted of numerous number of fibers: twisted, untwisted, twisted-and-untwisted	Resin reinforcement material for CFRP, CFRTTP or C/C composites, having such usage as Aircraft/Aerospace equipment , sporting goods and industrial equipment parts
Tow	An untwisted bundle of yarn constituted of extremely numerous number of fibers	Resin reinforcement material for CFRP, CFRTTP or C/C composites, having such usage as Aircraft/Aerospace equipment , sporting goods and industrial equipment parts
Staple Yarn	A yarn made of spinning of staples	Heat Insulator, Anti-friction material, C/C composite parts
Woven fabric	A woven sheet made of filament or staple yarn	Resin reinforcement material for CFRP, CFRTTP or C/C composites, having such usage as Aircraft/Aerospace equipment , sporting goods and industrial equipment parts
Braid	A braided yarn made of filament or tow	Resin reinforcement material particularly suitable for reinforcement of tubular products
Chopped fiber	A chopped fiber made of sized or non-sized fiber	Compounded into plastics/resins or portland cement to improve mechanical performances, abrasion characteristic, electric conductivity and heat resistance
Milled	Powder made by milling fiber in a ball-mill etc.	Compounded into plastics/resins or rubber to improve mechanical performances, abrasion characteristic, electric conductivity and heat resistance
Felt.Mat	A felt or mat made by layering up of staple by carding etc. then needle-punched or strengthened by organic binders	Heat insulator, base material for molded heat insulator, protective layer for heat resistance and base material for corrosion-resisting filter
Paper	A paper made from staple by dry or wet paper-making	Anti-electrostatics sheets, electrodes, speaker-cone and heating plate
Prepreg	An intermediate material in a form of half-hardened sheets made of Carbon Fibers impregnated with thermo-setting resin, qualities of which being stable and sustained long enough and therefore easily applicable for automatic sheet-layering	Aircraft/Aerospace equipment , sporting goods and industrial equipment parts needing lightness in weight and high performances
Compounds	A material for injection moulding etc. made of mixture of thermo-plastics or thermo-setting resins added by various additives and chopped fiber and then being compounded	Housing etc. of OA equipment taking advantages of electric conductivity, rigidity and lightness in weight

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Carbon Fiber Products

Type	Modulus	Filament Symbol	Number of Filaments	Mean elastic Modulus (GPa)
<i>Convencional tows</i>	Standard	3K	3000	230
		6K	6000	
		12K	12000	
		24K	24000	
	intemEDIATE	6K	6000	300
		12K	12000	
		24K	24000	
	High	6K	6000	350
		12K	12000	
		24K	24000	
<i>Large tows</i>	standard	45K	45000	230
		48K	48000	
		80K	80000	

Fiber bundle dimensions

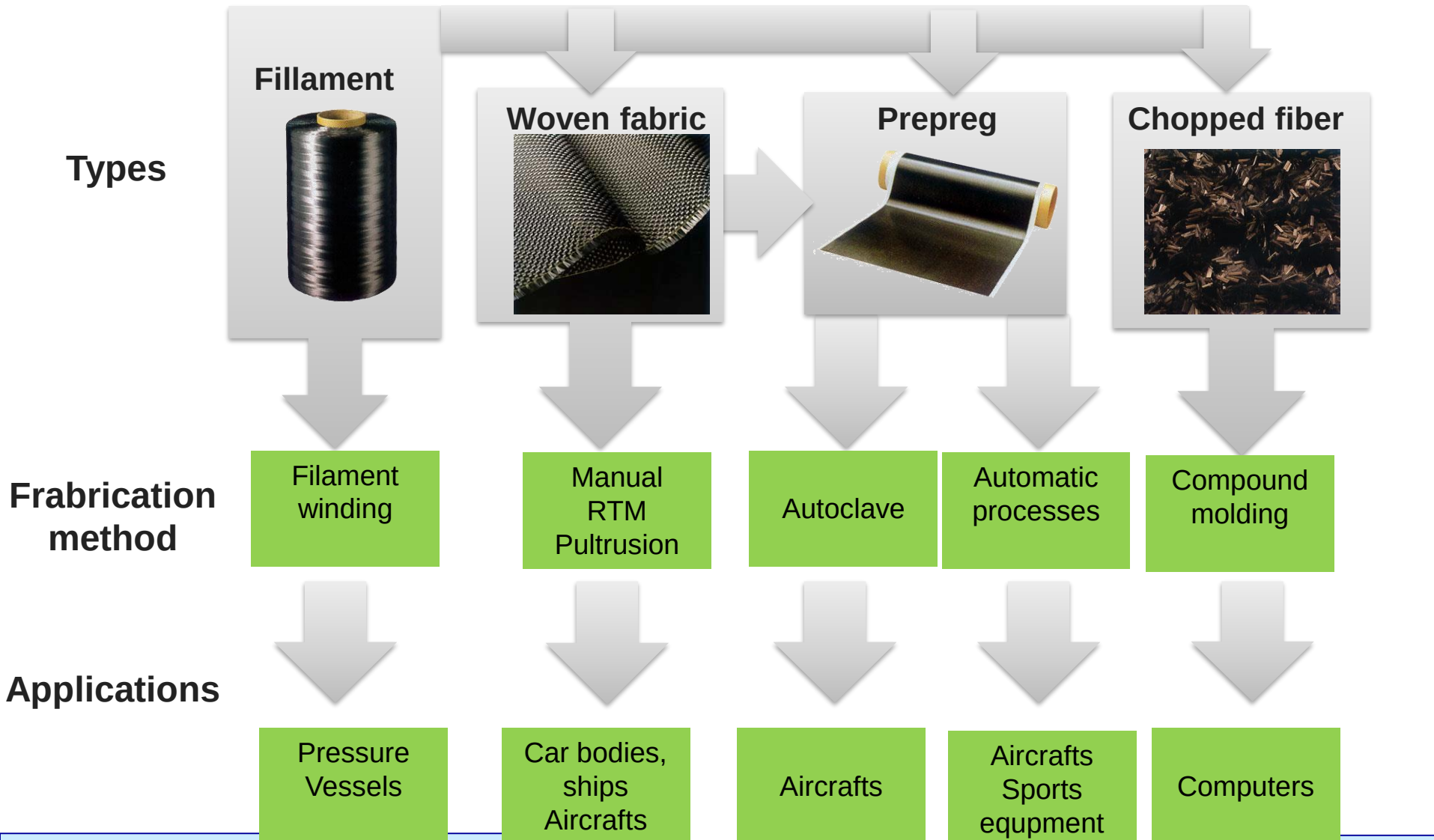
Material	Yield/tow		Filament size	
	m/kg	yd/lb	µm	µin.
Graphite (1000 to 12,000 filaments per tow)	300–1200	150–600	5–10	200–390
Fiberglass (2450–12,240 filaments per tow)	490–2400	245–1200	4–13	160–510
Aramid (800–3200 filaments per tow)	2000–7850	980–3900	12	470

Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

CARBON FIBERS

Fabrications methods and applications for the different types of **carbon fiber** products



Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

Polyamide: Kevlar or Nomex (poly(para-phenylene terephthalamide))

- **Aromatic ring** provides thermal stability
- **E ↑↑ due to its configuration:** rigid molecules are arrayed in ordered domains → during extrusion they are oriented in the direction of the flow
- Thermal and electrical insulator, $\downarrow \alpha$, high impact strength and $\downarrow E$ (compared to carbon)

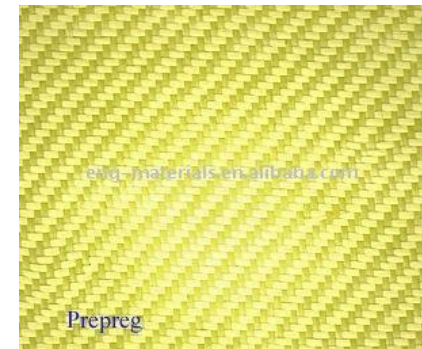
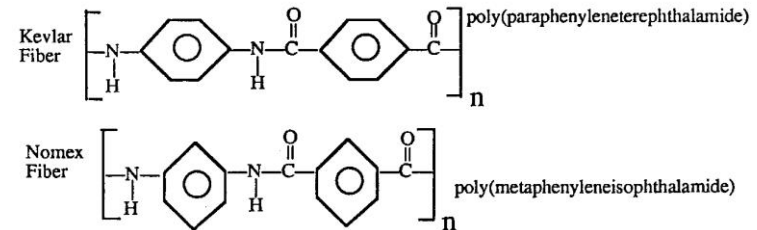
Types of Kevlar fibers (commercially introduced in 1972 by Du Pont):

Kevlar 49 → most used structural composite due to its $\uparrow E$

Kevlar 29 → high toughness applications (i.e.: bulletproof jacket)

Kevlar 149 → value of $E \approx$ theoretical $\uparrow \uparrow \uparrow E$

ARAMID FIBERS



100% Aramid fibre prepreg



Properties of the three types of Kevlar

Material	ρ (g/cm ³)	D_{wire} (μm)	σ_{tensile} (GPa)	E (GPa)	ϵ (%)
Kevlar 29	1,44	12	3,6	83	4,0
Kevlar 49	1,44	12	3,6-4,1	131	2,8
Kevlar 149	1,47	12	3,4	186	2,0

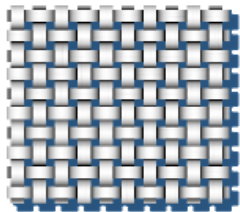
Fiber-Reinforced Composite Materials

FIBERS USED TO REINFORCE PLASTIC MATRICES

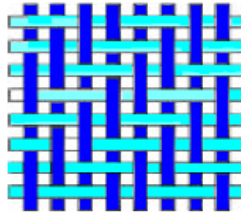
Woven fiber fabric:

CARBON, GLASS, ARAMID, and other multi-filament fibers are produced in **various fabric forms** for processing into high-performance PMCs using pre-preg, resing film or liquid resin injection techniques

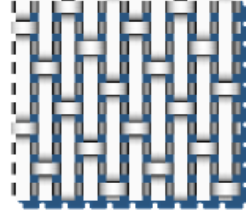
- **UND (uni-directional)**: cloth has 95% of the fibers woven parallel to the selvage, giving exceptional strength in that direction and very little at right angles to it
- **BID (bi-directional)**: cloth has half of the fibers woven parallel to the selvage edge of the cloth and the other half at right angles to the selvage, giving the cloth the same strength in both directions



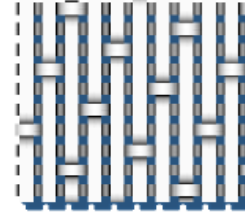
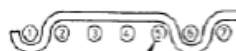
Plain weave



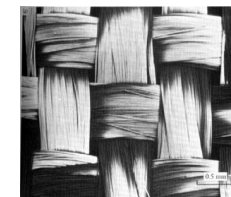
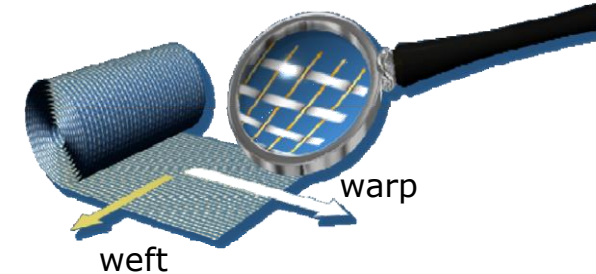
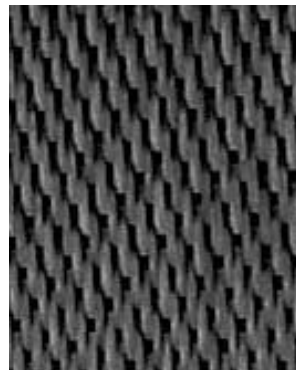
2x2 twill



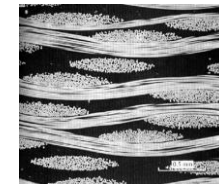
4 HS



8 HS
(Harness Satin)



Woven fibers



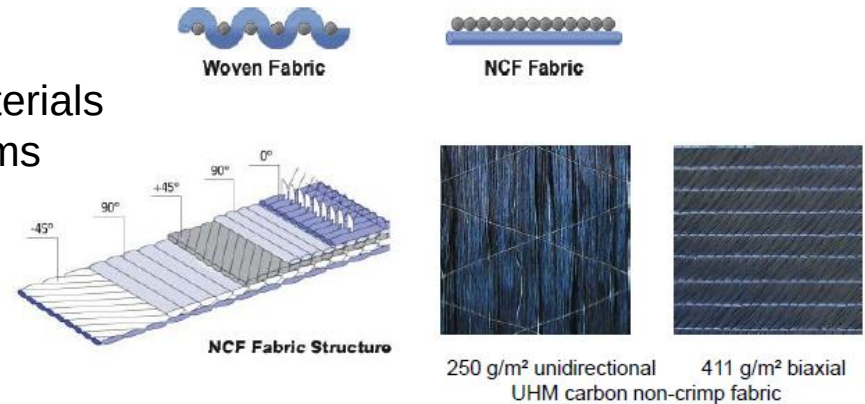
Cross-section view

Fiber-Reinforced Composite Materials

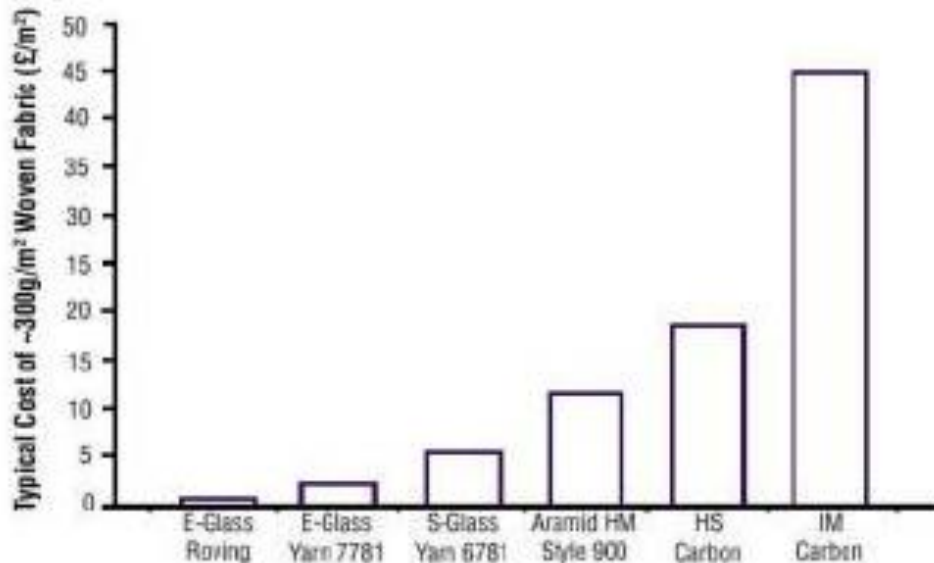
FIBERS USED TO REINFORCE PLASTIC MATRICES

Woven fiber fabric:

- Non Crimp Fabrics (NCFs)** : stitch-bonded materials with zero elongation made of unbent fiber systems



Typical cost of woven fabric

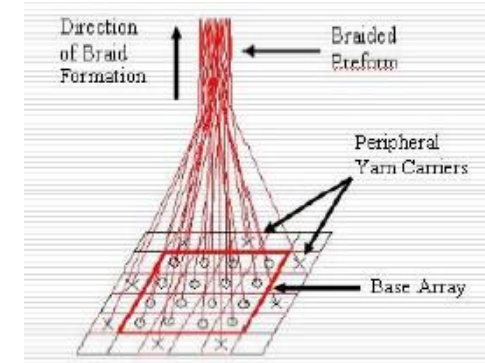


Fiber-Reinforced Composite Materials

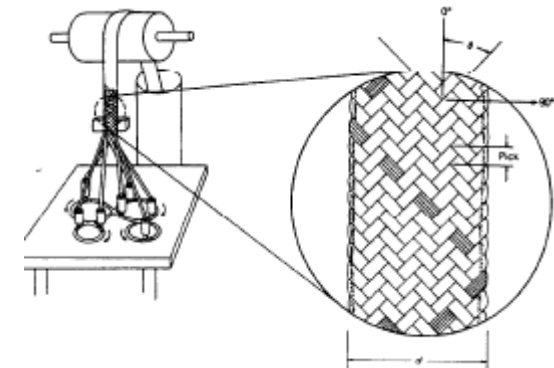
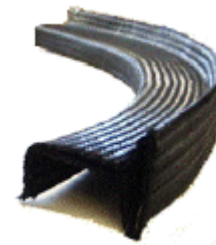
FIBERS USED TO REINFORCE PLASTIC MATRICES

PREFORMS

- Preforms are a set of dry fiber having very nearly the final shape of the part to be produced.
- Can be prepared by manual or automated methods.
- Textile technologies are used to produce dry fiber forms



3D braiding



3D carbon fiber preforms

Fiber-Reinforced Composite Materials

CERAMIC FIBERS

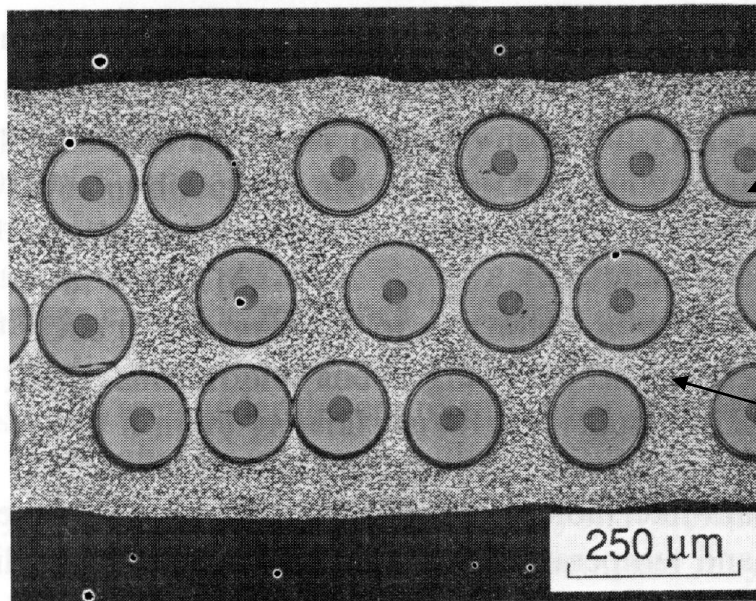
Mainly Boron, Al_2O_3 , Si_3N_4 , SiC ...

⇒ Typical matrices: *Al, Ti ...*

⇒ Properties: *Can resist high T ($T > 1300^\circ\text{C}$) and thermal shocks*

⇒ Applications: *Thermal insulator. Not structural applications.*

⇒ Limitation: *very expensive (5 times the price of carbon fiber)*



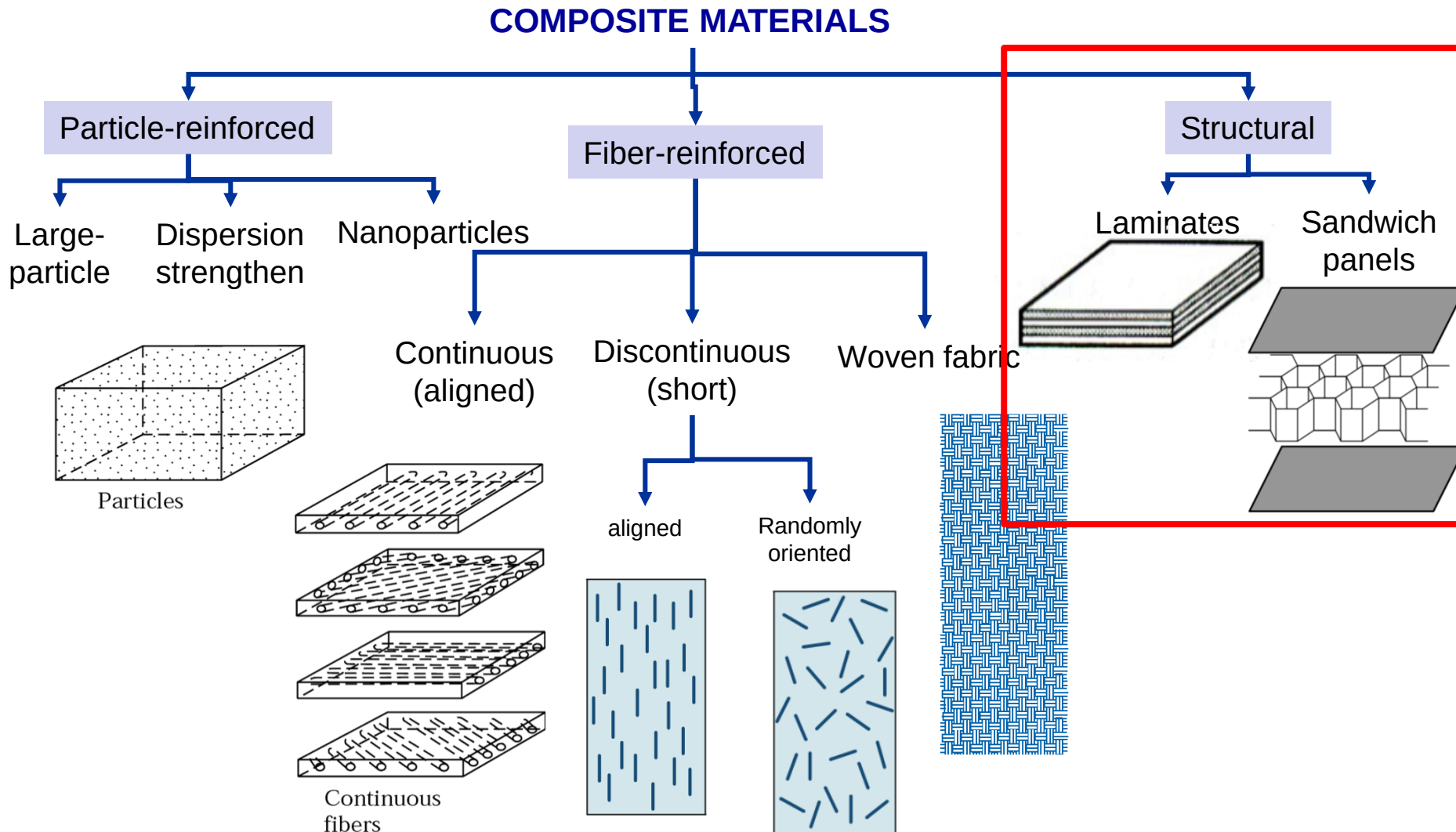
Carbon Coated
SiC Fibres

**SiC fibre-reinforced
Titanium Aluminide (MMC)**

Ti_3Al
Matrix

Classification of composite materials

According to the **reinforcement**:



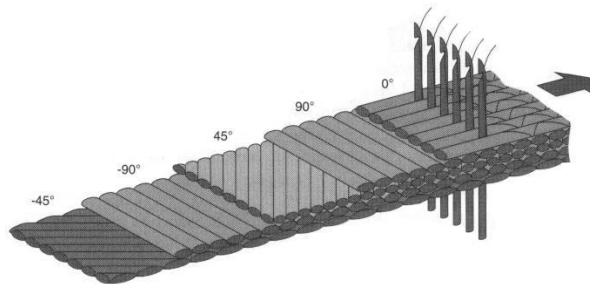
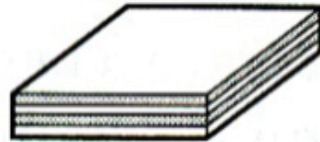
Structural Composite Materials

STRUCTURAL COMPOSITE MATERIALS

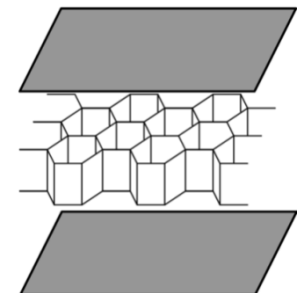
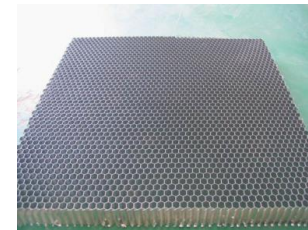
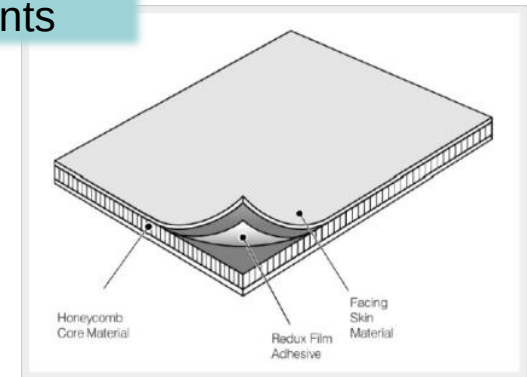
Laminates



- Formed by composite materials and homogenous materials
- The properties depend on the geometry of the structural elements



Sandwich Panels



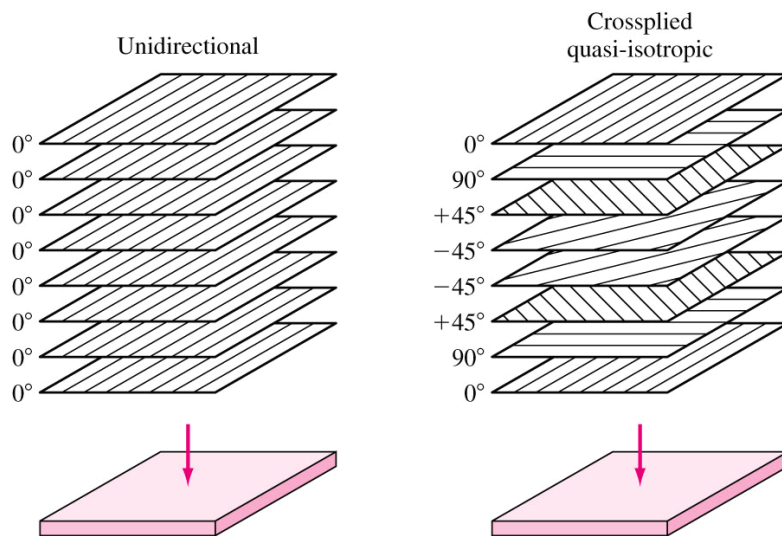
Structural Composite Materials

Laminates

Stacking sheets of unidirectional CM → In order to obtain different mechanical characteristics ⇒ stack laminates in different orientations with respect to each other.

Examples of laminates: plastics reinforced with continuous fibers and aligned with the matrix such as epoxy, polyester PE, PA, PET...

- Usual fiber orientations: 0, 90, +/-45. Combining these orientations one can obtain the strength and stiffness in any orientation. **Plane isotropy** can be achieved.
- Fiber layers arranged in a way so that strength is maximized and weight is minimized



- Laminates **should always be symmetric** with respect to their midplane, and balanced to avoid anomalous distortions of the structure
- The strength and stiffness changes dramatically with orientation
- A stack of fabrics, **without** a connecting **ligament**, **does not have a structural use**. It therefore requires a matrix.
- **Unidirectional composites are never used**

Piled laminas. The orientation of the direction changes in each of the layers.

Structural Composite Materials

Sandwich

2 external strong layers (face sheets) attached to a layer of less dense material (core) with low stiffness and low strength

- **Function of the face sheets** they withstand most of the plane loads and transversal bending stresses

Face sheet material Al alloys, fiber-reinforced plastics, Ti, steel and plywood.

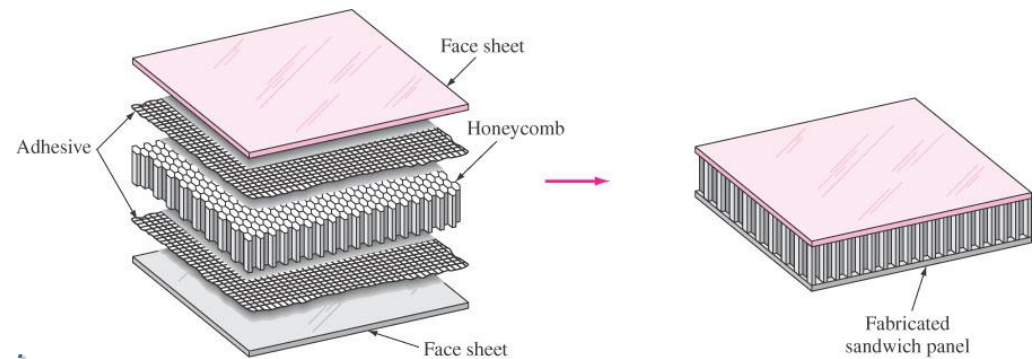
- **Function Core material** separates both face sheets and resists deformations $\perp$ to the face plane. Provide resistance to shear stress along the planes $\perp$ to the face sheets

■ **Core materials may have different and have different structures:** polymer foams, synthetic rubber, inorganic cement and balsa wood

■ **Typical core with honeycomb structure** → thin layers arranged in hexagonal cells.

Applications ⇒ *ceilings, floors and walls in buildings, in the aerospace industry (wing coating, fuselage)*

Processing of a sandwich panel with a honeycomb core



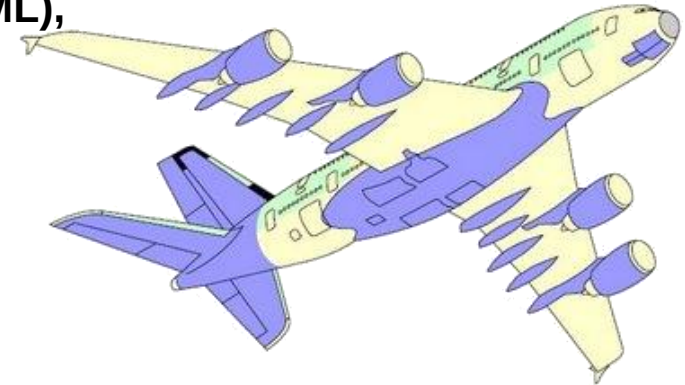
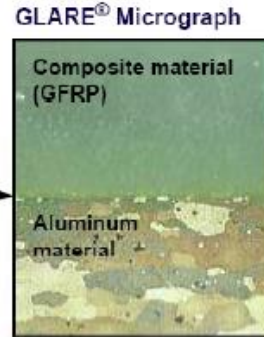
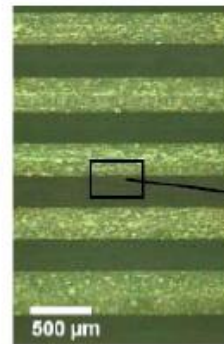
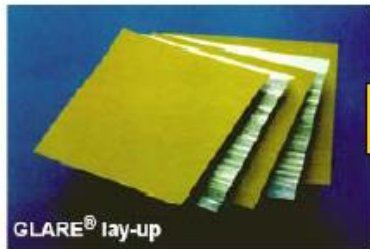
Structural Composite Materials

Sandwich

EXAMPLE

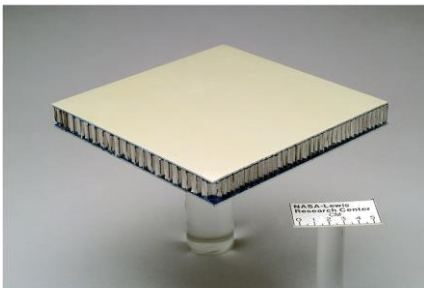
- Carbon Fiber Reinforced Plastic (CFRP)
- Glass Fiber Reinforced Plastic (GFRP)
- Quartz Fiber Reinforced Plastic (QFRP)
- Glass Reinforced Aluminum Laminate (GLARE)

GLARE "GLAss-REinforced" Fibre Metal Laminate (FML),

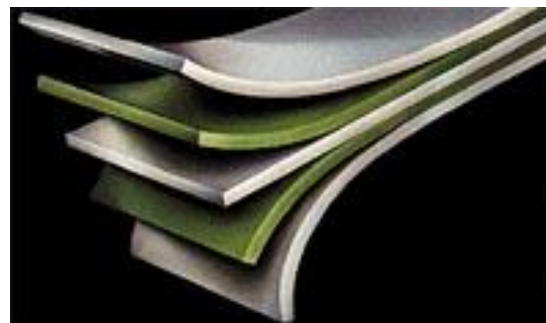


GLARE is a "glass-reinforced" Fibre Metal Laminate (FML), composed of several very thin sheets of metal (usually aluminum) intercalated with layers of "prepreg" fiberglass, all linked together with an epoxy matrix. Sheets of unidirectional prepregs may be aligned in different directions, looking for a suitable mechanical behavior to stress conditions provided.

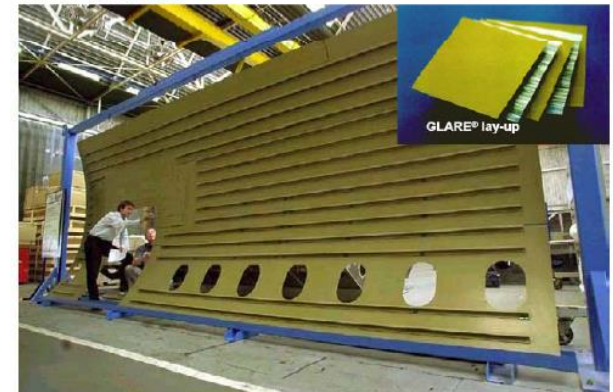
NASA
C-2000-1447



National Aeronautics and Space Administration
John H. Glenn Research Center at Lewis Field

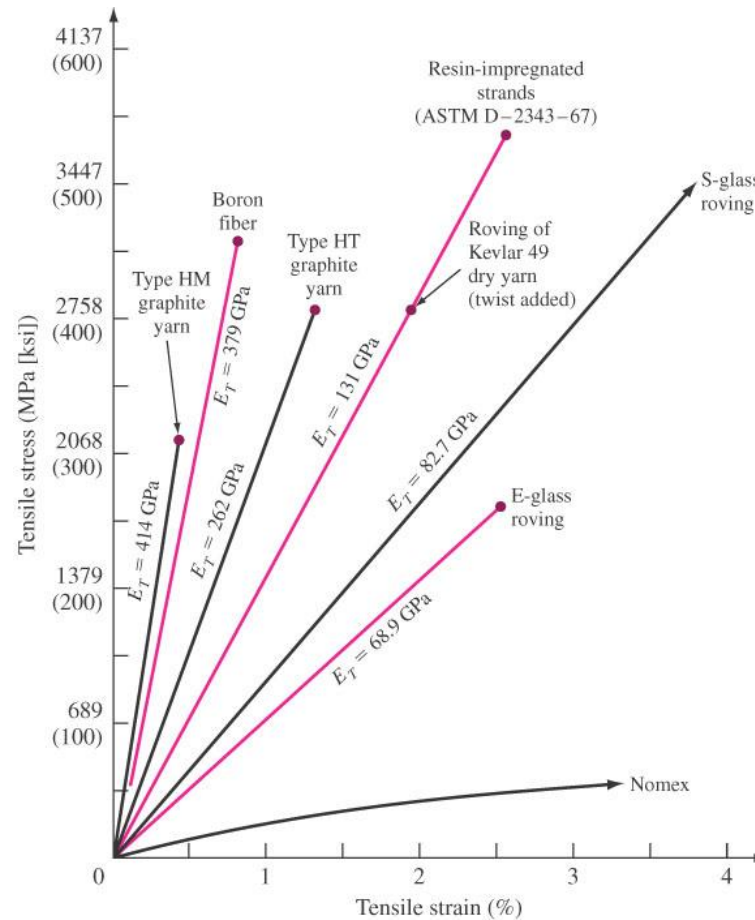


Airbus - Nordenham



Mechanical properties common reinforcement materials

Mechanical properties of the different fibers



Mechanical properties common reinforcement materials

Mechanical properties common epoxy PMC materials reinforced with carbon, aramid and glass fibers

	UD Eglass Epoxi	UD Carbon - epoxi	UD Kevlar49 -epoxi	0/90 Woven Eglass- epoxi	+/-45 woven Eglass- epoxi	0/90 Woven Carbon- epoxi	+/-45 woven Carbon- epoxi	0/90 Woven Kevlar- epoxi	CSM Eglass polyeste
% Vol fibra	53	57	60	33	33	50	50	50	19
UTS (0°) M Pa	1190	2040	1379	360	185	625	240	517	108
UTS (90°) M Pa	73	90	30	360	185	625	240	517	108
UCS (0°) M Pa	1001	1000	276	240	122	500	200	172	148
UCS (90°) M Pa	159	148	138	205	122	500	200	172	148
USS (0°) M Pa	67	49	60	98	137	130		110	85
E ₁ (0°) GPa	39	134	76	17	10	70	18	31	8
E (90°) GPa	15	11	5	17	10	70	18	31	8
Mod Cizalla G, GPa	4	5	2	5	8	5	27	2	2,75
ILSS (M Pa)	90	94	83	60	48	57	57	70	
Coef. Poisson		0,263	0,34	0,24	0,7				0,32
Densidad (g/cm3)	1,92	1,57	1,38	1,92	1,92	1,53	1,53	1,33	1,45

Mechanical properties common reinforcement materials

TABLE 10.11 MECHANICAL PROPERTIES OF COMMON DISPERSED-PHASE MATERIALS

Group	Dispersed phase	E [MPa (ksi)]	T.S. [MPa (ksi)]	Compressive strength [MPa (ksi)]	Percent elongation at failure
Glass fiber ^a	C-glass	69×10^3 (10×10^3)	3100 (450)	—	4.5
	E-glass	72.4×10^3 (10.5×10^3)	3400 (500)	—	4.8
	S-glass	85.5×10^3 (12.4×10^3)	4800 (700)	—	5.6
Ceramic fiber ^a	C (graphite)	$340 - 380 \times 10^3$ ($49 - 55 \times 10^3$)	2200–2400 (320–350)	—	—
	SiC	430×10^3 (62×10^3)	2400 (350)	—	—
Ceramic whisker ^a	Al ₂ O ₃	430×10^3 (62×10^3)	21×10^3 (3000)	—	—
Polymer fiber ^b	Kevlar ^c	131×10^3 (19×10^3)	3800 (550)	—	2.8
Metal filament ^a	Boron	410×10^3 (60×10^3)	3400 (500)	—	—
Concrete aggregate ^d	Crushed stone and sand	$34 - 69 \times 10^3$ ($5 - 10 \times 10^3$)	1.4–14 (0.2–2)	69–340 (10–50)	—

^aSource: Data from L. J. Broutman and R. H. Krock, Eds., *Modern Composite Materials*, Addison-Wesley Publishing Co., Inc., Reading, Mass., 1967.

^bSource: Data from A. K. Dhingra, Du Pont Company.

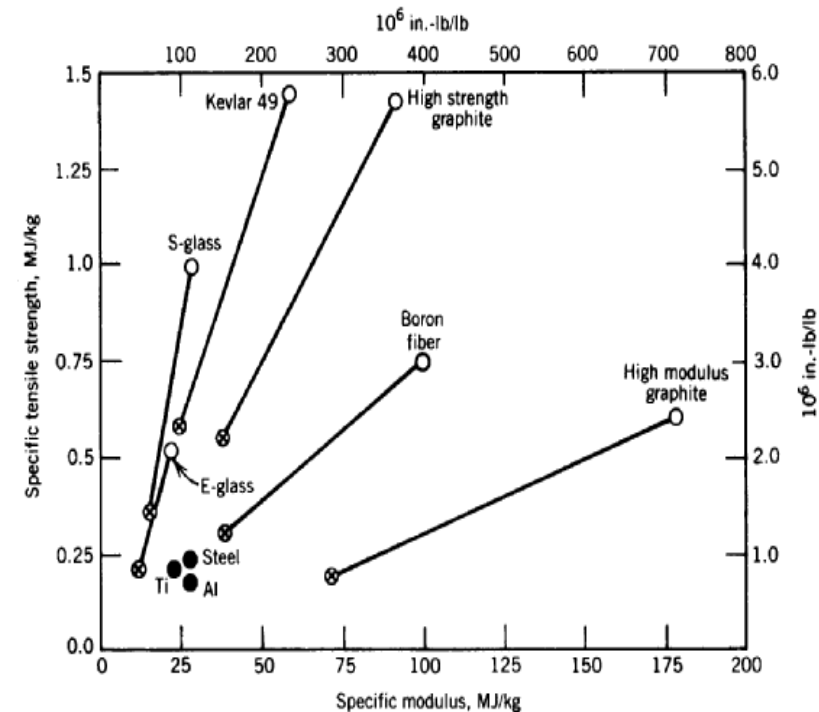
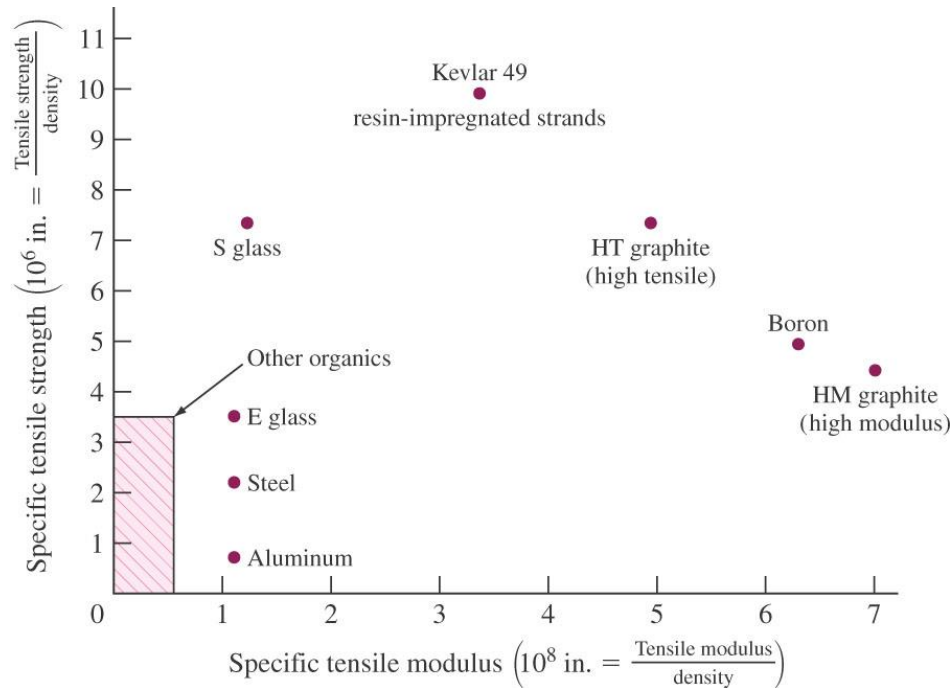
^cTrade name, Du Pont.

^dSource: Data from R. Nicholls, *Composite Construction Materials Handbook*, Prentice-Hall, Inc., Englewood Cliffs, N.J., 1976.

Mechanical properties common reinforcement materials

- Comparison of mechanical properties in terms of specific properties.

⊗: quasi-isotropic
I: bulk
○: unidirectional



Specific tensile strength and tensile modulus for different fibers in an epoxy matrix